MAINTENANCE CASE FILE — PART 1: DIAGNOSIS

INDEL M3.6 — ACT05 — Sistema de Gestión de Cafetería "Coffee House"

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Grade & section / Grado y sección	Bachillerato Técnico Vocacional en Desarrollo de Software — Sección A (please confirm)
System name / Nombre del sistema	Sistema de Gestión de Cafetería "Coffee House"
Stack / Tecnología	ASP.NET Web Forms (closest template category: Razor / ASP.NET) — C#, .NET Framework 4.7.2. Note: this is classic ASP.NET Web Forms, not Razor Pages/MVC and not ASP.NET Core, so it uses Web.config instead of appsettings.json.
Current version / Versión actual	v1.0 — no formal version numbering or Git tags were found in the delivered copy (finding F-08).
Date / Fecha	September 11, 2026 (adjust to your actual submission date)

1 · Configuration inventory

Inventario de configuración — based on a direct read of the delivered source code (Web.config, Conexion.cs, CafeteriaDB.sql and the .aspx.cs files). Fields marked PENDING require you to check the live system and cannot be answered from the code alone.

Item	Description (English)
Purpose of the system and users	Point-of-sale and inventory management system for a small cafeteria ("Coffee House"). It manages product categories, products and stock, clients, employees and sales, including an itemized cart-style checkout. Intended users are cafeteria staff (cashiers/administrators) who log in to register sales and maintain the catalog. *PENDING — confirm how many real users use it today versus only the project team.*
Language, framework and versions	C# on ASP.NET Web Forms. Target framework: .NET Framework 4.7.2 (set in Web.config under system.web / httpRuntime).
Database engine and version	Microsoft SQL Server, accessed as a local instance: (localdb)\MSSQLLocalDB, Integrated Security. *PENDING — exact SQL Server engine build not recorded; run SELECT @@VERSION to confirm.*
Number of tables and approximate size	6 tables: Categorias, Productos, Clientes, Empleados, Ventas, DetallesVenta (renamed from Detalle_Ventas). *PENDING — database size in MB not yet measured; check with sp_spaceused or SSMS → Reports → Disk Usage.*
Three largest tables and their record counts	*PENDING — must be measured live with SELECT COUNT(*) on each table; do not estimate. By design, Productos, Ventas and DetallesVenta are the tables most likely to grow with use.*
Dependencies and external services	NuGet package: Microsoft.CodeDom.Providers.DotNetCompilerPlatform 2.0.1 (per packages.config). Framework references: System.Data.SqlClient (ADO.NET), System.Web.*, System.Configuration. No external services are called (no email provider, no payment gateway,

	no cloud SDK).
Where the system runs	Local development machine only, launched from Visual Studio against a LocalDB instance. Not deployed to any server or hosting; not reachable from outside the machine it runs on.
Configuration files (names only, no passwords)	Web.config (holds the "ConexionBD" connection string), Web.Debug.config, Web.Release.config, Cafeteria.csproj, packages.config.
Credentials custody (who knows what)	The database connection uses Integrated Security (Windows authentication), so there is no separate database password stored anywhere. *PENDING — the team has not formally documented who is responsible for this configuration if a member is absent.*
Location of the official source code	Delivered as Proyecto-Cafeteria--main.zip. No .git folder was included in the delivered copy, so version history cannot be verified from this file. *PENDING — confirm whether this ZIP is exported from a GitHub repository the team maintains, and who holds write access.*
Existing documentation	One Word document (Cafeteria.docx, also exported as Cafeteria.pdf) with an introduction and screenshots of the SQL table creation and a few screens. There is no README, no data dictionary, no installation guide and no user manual.
Personal data stored	Clientes: full name, phone, email. Empleados: full name, username, password, position, phone, email, hire date. This is real personal data of clients and staff. Critically, Empleados.contrasena stores the login password as plain text (VARCHAR(50), no hashing) — see finding F-01.

2 · Backup and restore evidence

None of these steps have been performed yet. Follow the Razor/ASP.NET row of the reference table in the activity: back up the full Cafeteria solution folder plus a database export, store the second copy somewhere other than this laptop, then restore the database under the name Cafeteria_DEV and confirm the project opens against it.

Item	File name and date	Size	Where it is stored
Source code backup	PENDING — e.g. Cafeteria_CODE_2026-09-11.zip	PENDING	PENDING (2nd location must differ from the working laptop)
Database export	PENDING — e.g. Cafeteria_DB_2026-09-11.bak	PENDING	PENDING
Second copy location	PENDING — USB drive / cloud drive / teammate's machine	—	—
Restore test performed	PENDING — restore into a database named Cafeteria_DEV	—	—

How the restore test was done and what proved it worked — English

PENDING — *Complete this after performing the restore described above: state which database you restored the backup into, and what you checked (e.g. "the Login screen and the Productos grid loaded correctly against Cafeteria_DEV").*

Prueba de restauración — Español

PENDING — *Completar después de realizar la restauración: indicar en qué base se restauró la copia y qué se verificó.*

How you would return the system to today's state — English

Suggested procedure, to confirm after the real test: (1) Stop the application. (2) Restore the dated database backup over Cafeteria, or into Cafeteria_DEV and repoint the connection string. (3) Replace the Cafeteria code folder with the code backup dated the same day. (4) Reopen the solution and confirm the Login screen and the Productos grid work with the restored data.

Procedimiento de reversión — Español

Procedimiento sugerido, a confirmar tras la prueba real: (1) Detener la aplicación. (2) Restaurar el respaldo de la base de datos fechado sobre Cafeteria, o en Cafeteria_DEV y apuntar la cadena de conexión ahí. (3) Reemplazar la carpeta de código Cafeteria con el respaldo de código de la misma fecha. (4) Reabrir la solución y confirmar que el Login y la grilla de Productos funcionan con los datos restaurados.

3 · Baseline metrics

Métricas de línea base — measure honestly before making any change; do not estimate.

Metric	Value BEFORE	How it was measured
Load time of the heaviest screen	*PENDING*	Time Productos.aspx or Ventas.aspx (both join tables and fill a DataSet) with the browser Network tab or a stopwatch; repeat 3 times and average.
Time of the slowest query or report	*PENDING*	Time the product search (txtBuscar_TextChanged, a LIKE query joining Productos and Categorias) from click to results.
Database size	*PENDING*	SSMS → right-click Cafeteria → Reports → Disk Usage, or run sp_spaceused.
Records in the main table	*PENDING*	SELECT COUNT(*) FROM Productos; repeat for Ventas and DetallesVenta.
Known open defects	8 identified so far (see Findings register, section 4)	Counted directly from the findings register below.
Repeated errors in the log	*PENDING*	The app has no dedicated log file configured; check the Visual Studio Output window / IIS Express console while reproducing common actions.
Active users	*PENDING*	Count real people who actually log in today, not the number of rows in Empleados.
Date of the last backup before today	Never	No backup of this system exists prior to this diagnosis. This absence is itself the finding (see F-08).

4 · Findings register

Registro de hallazgos — 8 findings identified from a direct code review, spanning Security, Database, Performance/code, and Users/documentation.

ID	Finding (English)	Area	Severity	Evidence / how detected	Maintenance type
F-01	Employee passwords are stored in plain text in Empleados.contrasena (VARCHAR(50), no hashing), and Login.aspx.cs compares them in plain text.	Security	**High**	Empleados table schema (CafeteriaDB.sql) and Login.aspx.cs, lines ~34-38.	Corrective
F-02	Login.aspx.cs uses parameterized SQL (good practice), but four other screens do not check the session before loading data.	Security	**High**	Categorias.aspx.cs, Clientes.aspx.cs, Empleados.aspx.cs, Productos.aspx.cs — no Session["Empleado"] check in Page_Load, unlike Inicio.aspx.cs.	Corrective
F-03	The database connection string is hardcoded and duplicated in four code files instead of always being read from Web.config through the existing Conexion class.	Performance and code	Medium	Categorias.aspx.cs, Clientes.aspx.cs, Empleados.aspx.cs, Productos.aspx.cs, line 11 in each: new SqlConnection(@"Data Source=(localdb)\MSSQLLocalDB...").	Preventive
F-04	Ventas.aspx.cs contains an empty catch block that silently swallows any exception, hiding real errors from anyone maintaining the system.	Performance and code	Medium	Ventas.aspx.cs, line 141: catch { }.	Corrective
F-05	The Empleados.usuario and Empleados.contrasena columns were altered to allow NULL, so an employee record can exist with no login credentials at all.	Database	Medium	CafeteriaDB.sql: ALTER TABLE Empleados ALTER COLUMN usuario VARCHAR(50) NULL / contrasena VARCHAR(50) NULL.	Corrective
F-06	No indexes exist beyond the primary keys; foreign keys are present, but there is no index review for the columns most used in searches (e.g. Productos.nombre_producto).	Database	Low	CafeteriaDB.sql — only PRIMARY KEY IDENTITY constraints are defined; no CREATE INDEX statements.	Preventive

F-07	There is no backup, no verified restore, and no separate development copy of the system as of this diagnosis.	Users and documentation	**High**	No .bak/.sql export or DEV database exists yet; see section 2 of this file (all PENDING).	Preventive
F-08	Documentation is limited to one Word/PDF report with screenshots; there is no README, no data dictionary and no version control history included in the delivered copy, so another developer could not install or run the system from what exists today.	Users and documentation	Medium	Delivered ZIP contains Cafeteria.docx/.pdf only; no .git folder, no README.md.	Preventive

Overall state of the system — English

The system correctly implements its core CRUD and sales workflow, and the Login screen uses parameterized SQL, which avoids classic SQL injection there. However, it is not yet safe to maintain as a live system: employee passwords are stored and compared in plain text (F-01), most screens do not check that a user is logged in before loading data (F-02), and the database connection string is duplicated by hand across four files instead of being read consistently from Web.config (F-03). No backup, restore test, development copy, or measured baseline exists yet. Before any new feature work, the team should close F-01 and F-02 first, since both are high-severity security gaps, and then complete the backup, restore and baseline steps left pending in this file.

Estado general del sistema — Español

El sistema implementa correctamente su flujo principal de CRUD y ventas, y la pantalla de Login usa SQL parametrizado, lo que evita la inyección SQL clásica en ese punto. Sin embargo, todavía no es seguro mantenerlo como sistema en producción: las contraseñas de empleados se guardan y comparan en texto plano (F-01), la mayoría de pantallas no verifica que haya una sesión iniciada antes de cargar datos (F-02), y la cadena de conexión a la base de datos está duplicada manualmente en cuatro archivos en vez de leerse siempre desde Web.config (F-03). Aún no existe respaldo, prueba de restauración, copia de desarrollo ni línea base medida. Antes de agregar funciones nuevas, el equipo debería cerrar primero F-01 y F-02, por ser hallazgos de seguridad de severidad alta, y luego completar los pasos de respaldo, restauración y línea base que quedan pendientes en este archivo.